

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

UNIT AND MEASUREMENTS

LECTURE-4

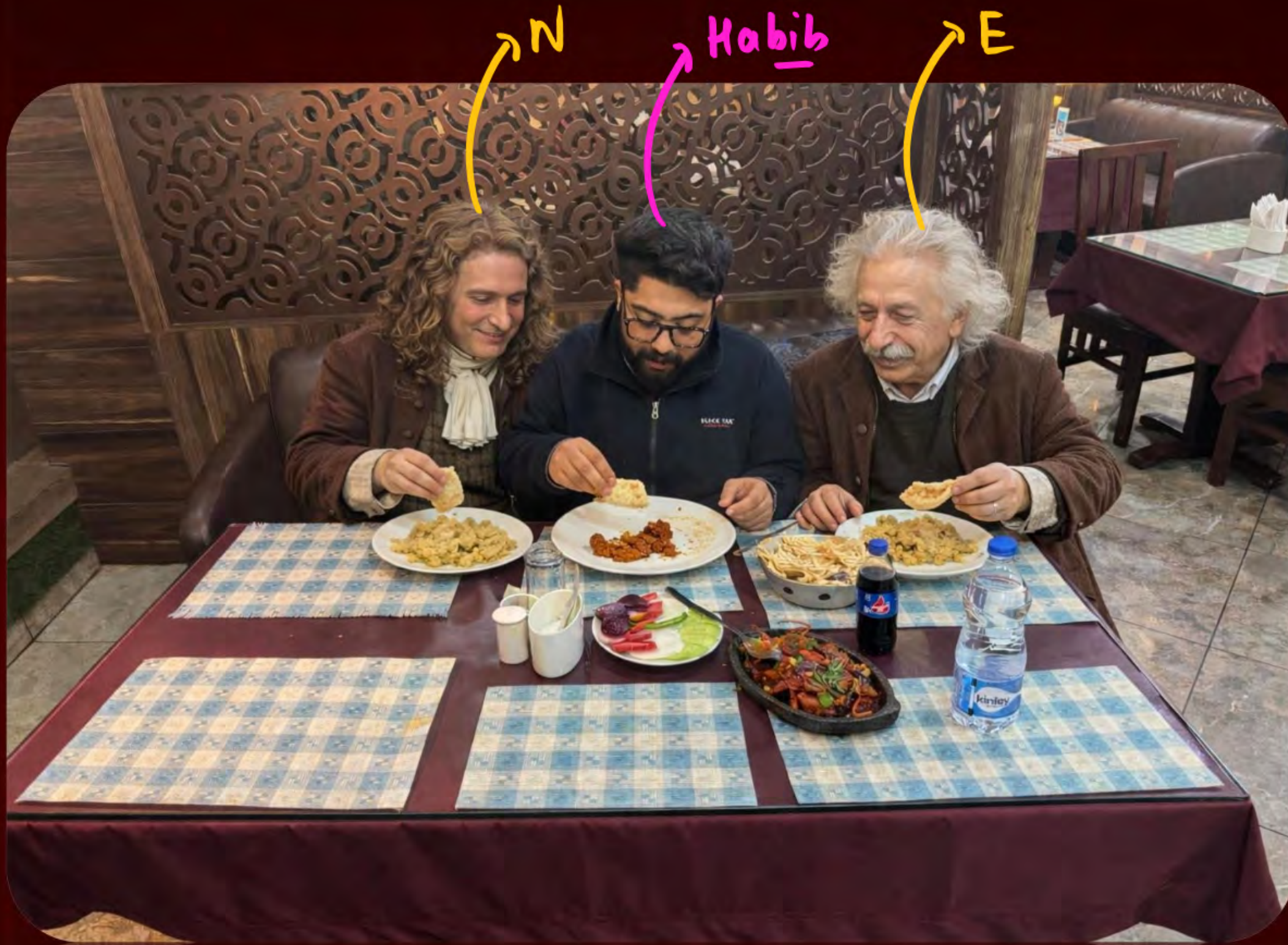
PHYSICS

WAJAHID^{Sir}

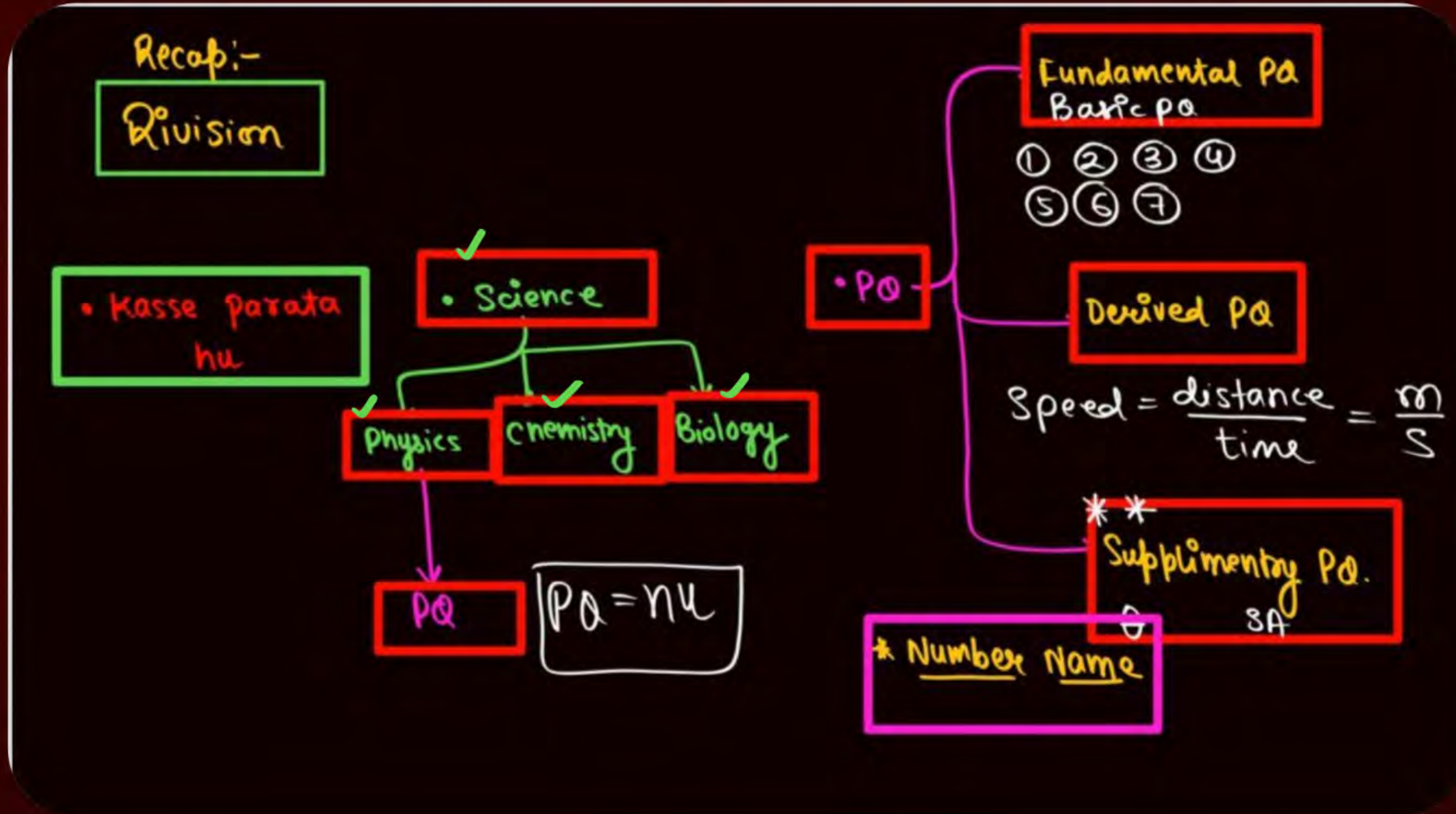


Topics To Covered

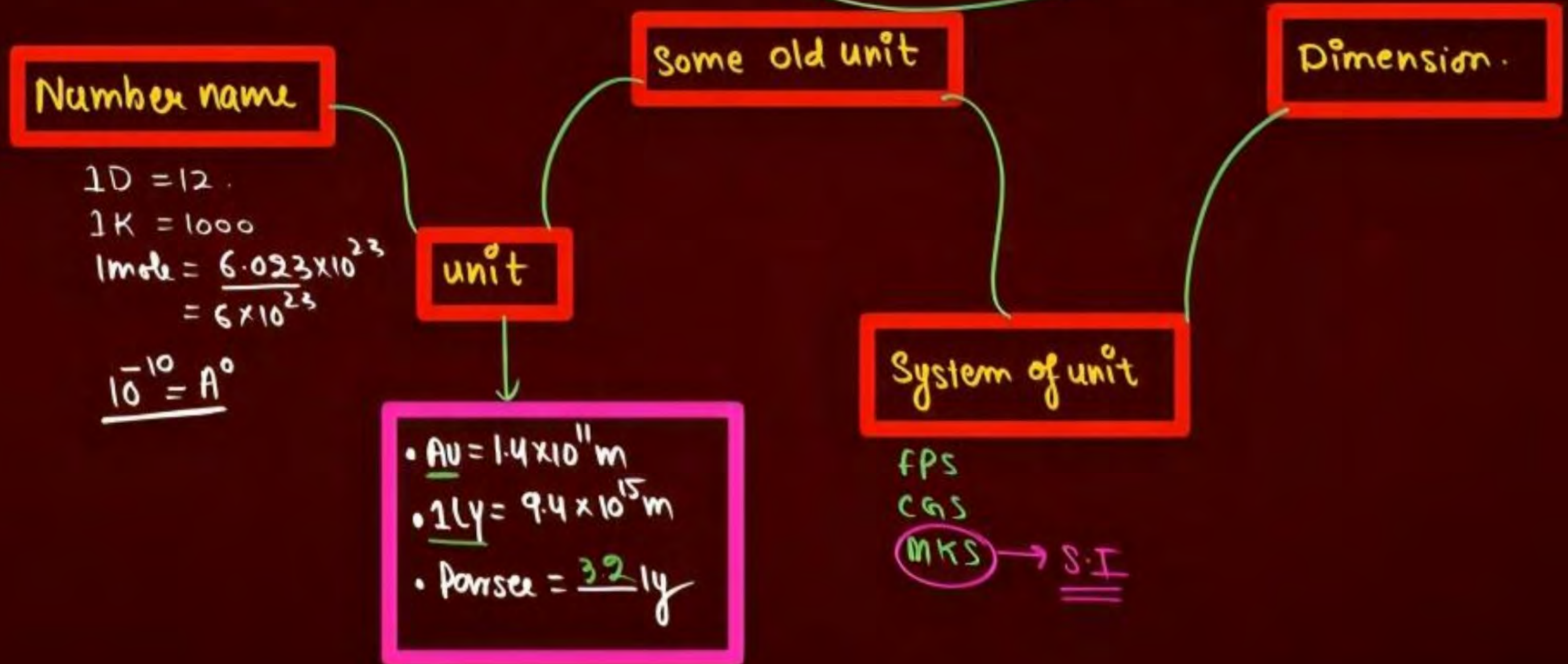
- 1 REVISION ✓
- 2 APPLICATION OF DIMENSION ✓
- 3 (a) TO FIND THE DIMENSIONAL FORMULA of any PQ
- 4 (b) TO CHECK THE CORRECTNESS OF PHYSICS FORMULA



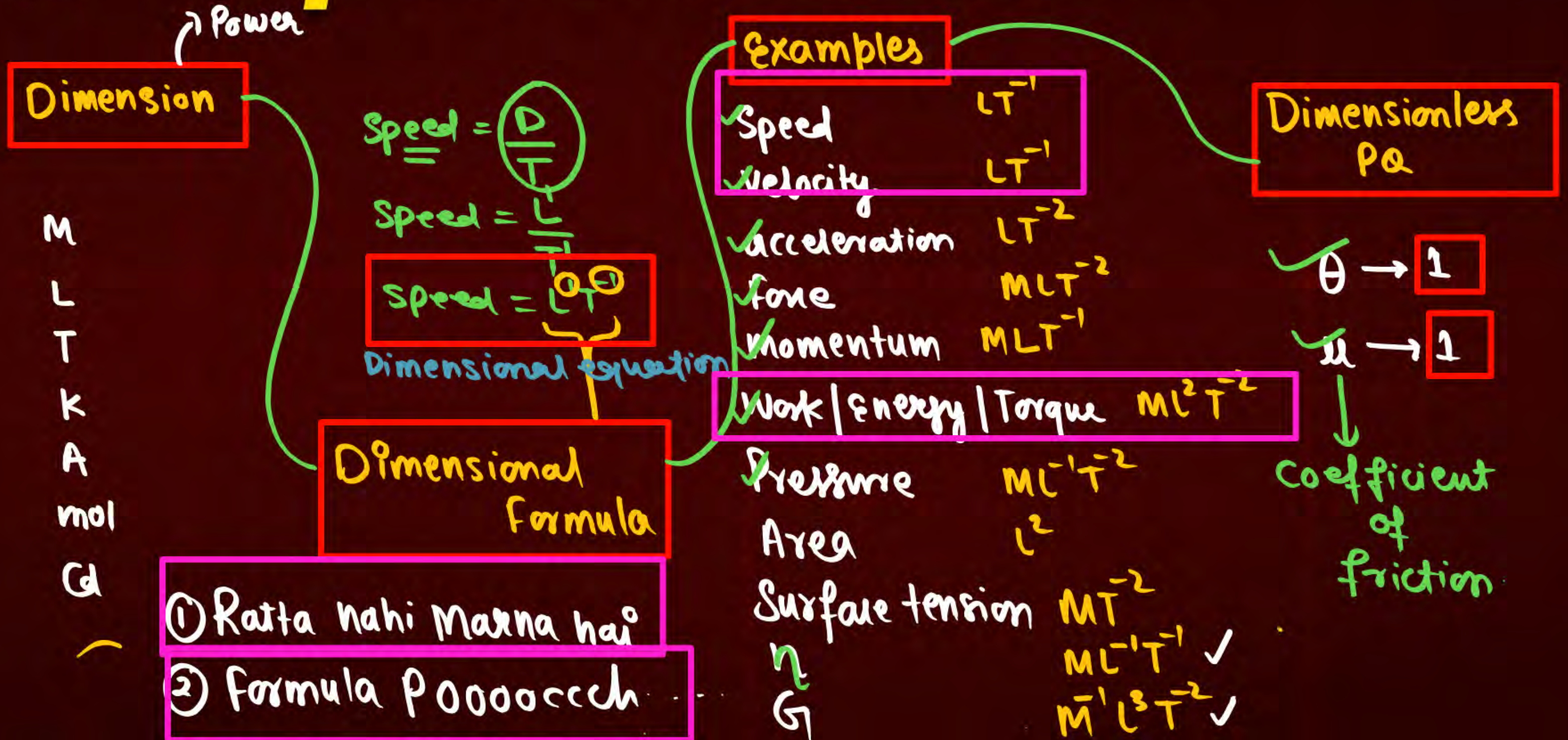
Recap From 1st Lec.



Recap From 2. Lec.



Recap From 3rd Lec.



Lorentz Force

centripetal force

Gravitational Force

Note

electro
motive
force

EMF

Dokhe baaz hai

MLT^{-2}

12th

V

Potential difference

Surface tension
 MT^{-2}

friction force

Magnetic force

Spring force

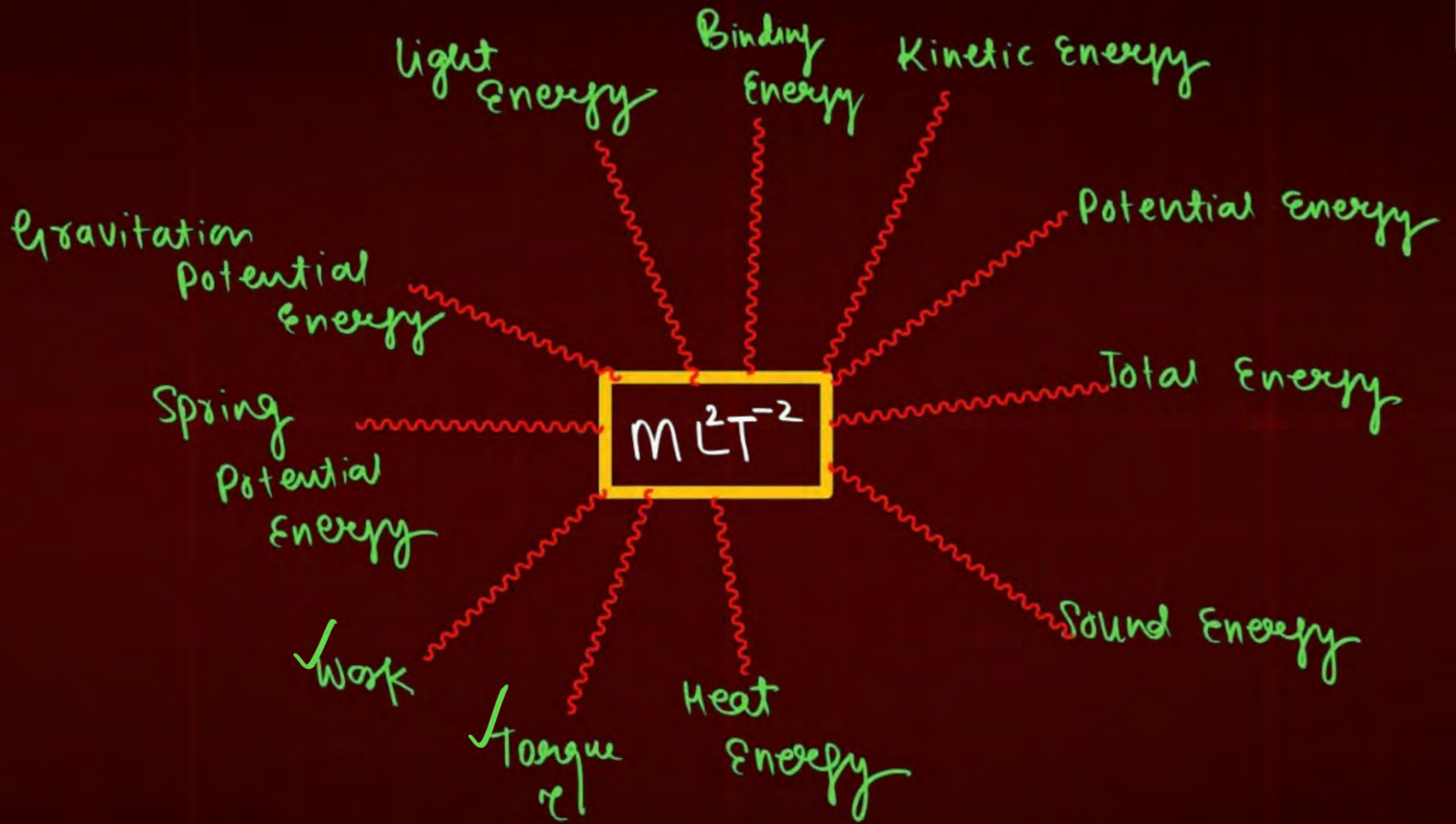
Tension
force

Electrostatic
force

Normal
force

centrifugal force

$F = MLT^{-2}$



Lecture. no. (4)

Application of Dimension Analysis

$$F_c = \frac{mv^2}{R}$$

- (a) To find out the Dimensional formula of physical Quantity G / η
- (b) To check the correctness of physical formula
(i) $v = u + at$ (ii) $s = ut + \frac{1}{2}at^2$ (iii) $v^2 = u^2 + 2as$
- (c) To derive physics formula. $T = 2\pi \sqrt{l/g}$
- (d) To convert one system of unit into another system of unit.

1) TO FIND DIMENSIONAL FORMULA of PQ



7. To find the Dimensional formula of G

$$\rightarrow F = \frac{G m_1 m_2}{r^2}$$

$$F r^2 = G m_1 m_2$$

$$\frac{F r^2}{m_1 m_2} = G$$

$$G = \frac{F r^2}{m_1 m_2}$$

$$G = \frac{MLT^{-2} L^2}{M M}$$

$$G = \frac{L^3 T^{-2}}{M}$$

$$G = M^{-1} L^3 T^{-2}$$

$$G = M^{-1} L^3 T^{-2}$$

$$G = M^{-1} T^{-2} L^3$$



Q. To Find the dimensional formula of $(\eta) \rightarrow$ Coefficient of viscosity
 and Stokes law $\rightarrow$ radius η

507

Stroke law

radium

Force $F_v = \underbrace{6\pi}_{\text{ignore}} \underbrace{\eta}_{\checkmark} r v \rightarrow \text{velocity}$

$$F_v = \eta \gamma \dot{v}$$

$$\frac{F}{\gamma v} = \eta$$

$$\eta = \frac{F}{V}$$

$$\eta = \frac{F}{\gamma v}$$
$$\eta = \frac{MKT^{-2}}{KLT^{-1}}$$

$$\eta = M C^{-1} T^{-1}$$

Q Dimensional formula of L (Angular momentum) → 2017

Sol

$$L = \overset{\text{radius}}{r} p$$

$$L = L' M L' T^{-1}$$

$$L = M L^2 T^{-1}$$

→ angular momentum

$$p = mv$$
$$p = M L T^{-1}$$

QUESTIONS

2019 (PYQ)

JKBOSE



A physical quantity is given by $h = \frac{Fv^2}{L}$ is the force, v is the velocity and L is the angular momentum. Find the dimensions of h .

1 ☒ [M⁰LT⁻³]

2 ☐ [MLT⁻²]

3 ☐ [M²L²T⁻⁴]

4 ☒ None of these

$$h = \frac{Fv^2}{L}$$

$$\bullet F = MLT^{-2}$$

$$v = LT^{-1}$$

$$L = \gamma p$$

$$\gamma \rightarrow L$$

$$p \rightarrow MLT^{-1}$$

$$h = \frac{MLT^{-2}(LT^{-1})^2}{L MLT^{-1}}$$

$$h = \frac{T^{-1}L^2T^{-2}}{L}$$

$$h = T^{-1}LT^{-2}$$

$$h = LT^{-1}T^{-2}$$

$$h = LT^{-3}$$

$$h = M^0L^1T^{-3}$$

$$h = L T^{-3}$$

$$h = m^0 L' T^{-3}$$

Q Find the dimensional formula of Planck's constt (h) (Neet - 2019)

sol $\boxed{E = h\nu}$ frequency $\nu = \frac{1}{T}$ frequency = $\frac{1}{T}$

$$E = h \times \frac{1}{T}$$

$E \times T = h$

Energy
↓
Time
Period

$$h = E \times T$$

$$h = M L^2 T^{-2} \times T$$

$$h = M L^2 T^{-2+1}$$

$$\boxed{h = M L^2 T^{-1}}$$

Q Find the dimensional formula of Refractive index (μ)

Sol $\mu = \frac{c}{v}$

$c \rightarrow$ Speed of light in vacuum

$v \rightarrow$ Speed of light in medium

$$\mu = \frac{[L T^{-1}]}{[L T^{-1}]}$$

$\boxed{\mu = 1}$ $\rightarrow$ Dimensionless quantity

2) TO CHECK THE CORRECTNESS OF PHYSICS FORMULA

Newton chacha.

Law of Homogeneity

$$1\vec{v} + 1\vec{v} = 2\vec{v}$$

$$2\vec{v} + 1\vec{v} = 3\vec{v}$$

$$2\vec{v} - 1\vec{v} = 1\vec{v}$$

$$3\vec{v} - 1\vec{v} = 2\vec{v}$$

$$\bullet x + x = 2x$$

$$\bullet 2x - x = x$$

Same PQ added, Subtracted and equated

$$\# \underline{3\text{m/s}} - \underline{1\text{m/s}} = 2\text{m/s}$$

$$\# \underline{2\text{N}} + \underline{1\text{N}} = 3\text{N}$$

$$\times \# \underline{2\text{m/s}} + \underline{1\text{N}} \neq 3$$

$$\times \# \underline{3\text{m/s}} - \underline{1\text{N}} \neq 2$$



$$\begin{array}{c}
 \text{apple} + \text{banana} \neq 2\text{apple} \\
 x + y \neq xy \quad \text{apple} \neq \text{banana}
 \end{array}$$

$$\begin{array}{c}
 \text{apple} - \text{banana} \neq \cancel{\text{apple}} \\
 x - y \neq x
 \end{array}$$

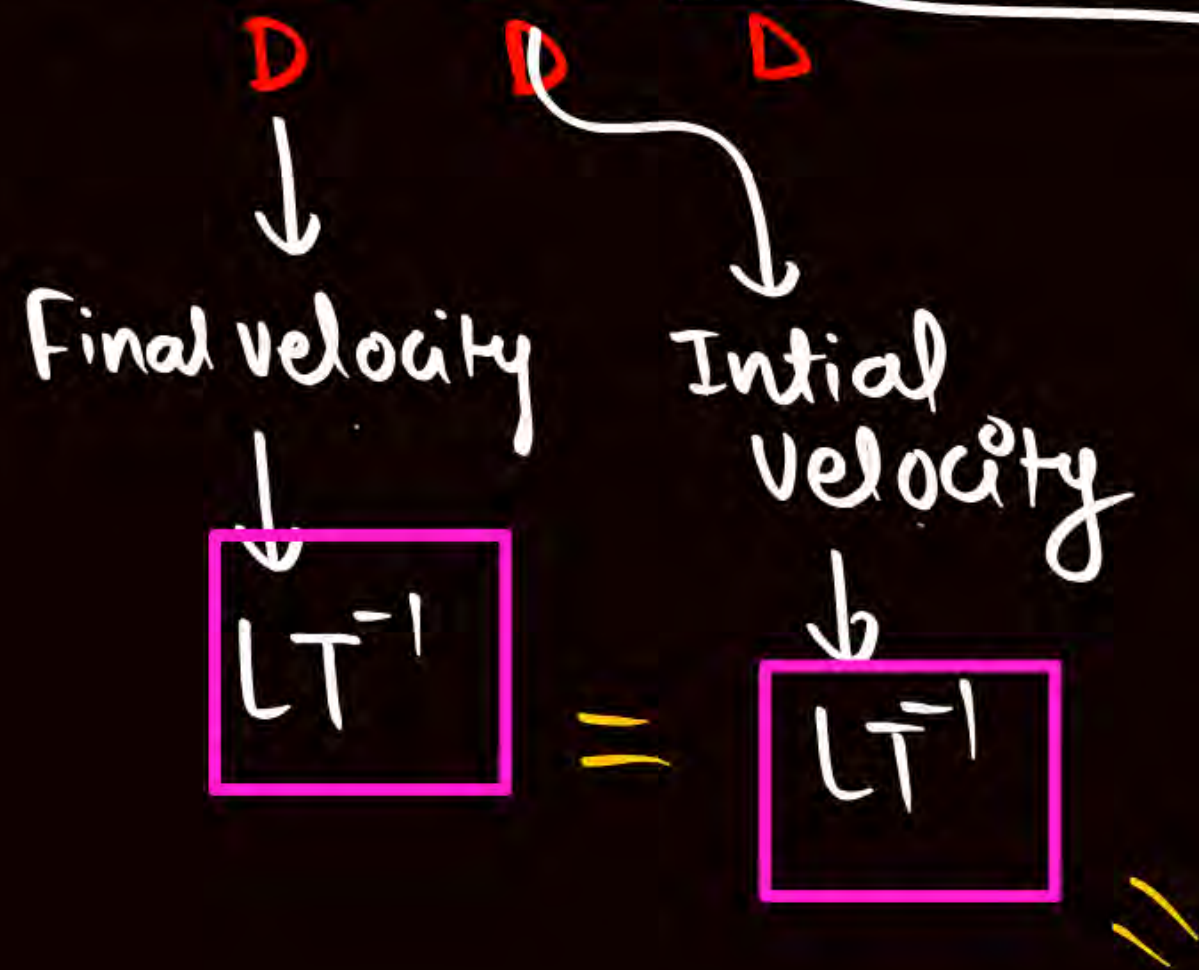
$$1gf + 1gf = 2gf$$

$$2gf - 1gf = 1gf$$

$$2w + 1gf \neq 3$$

$$2w - 1gf \neq 1$$

(1) $V = U + at$ (qth equation of motion)



at

$LT^{-2}T^1$

LT^{-2+1}

LT^{-1}

All PQ $\rightarrow$ U/V and at have same dimensional formula

hence $V = U + at$ is correct

$$(ii) \quad \underbrace{S}_D = \underbrace{ut}_D + \underbrace{\left(\frac{1}{2}\right)at^2}_D$$

2020 (PYQ)

JKBOSE

$$\underline{2N} + \underline{1N} = 3N$$

$S \rightarrow$ Displacement

$t \rightarrow$ time

$$\underline{2N} + \underline{1m/s} \neq 3$$

$u \rightarrow$ initial velocity

$a \rightarrow$ acceleration

$$\boxed{S} = \boxed{ut} + \boxed{\left(\frac{1}{2}\right)at^2}$$

$\downarrow$
 $\downarrow$
 $\downarrow$

$$LT^{-1}T^1$$

$$\downarrow$$

 LT^{-1+1}

$$= LT^0 = |x| = L = \frac{|x|}{1}$$

$$LT^{-2}T^2$$

$$\downarrow$$

 $L T^{-2+2}$

$$LT^0$$

(iii) $V^2 = U^2 + 2aS$

$(LT^{-1})^2 = (LT^{-1})^2 + 2 \frac{L}{T^2} L$

$L^2 T^{-2} = L^2 T^{-2} = L^2 T^{-2}$

$V \rightarrow$ Final velocity

$u \rightarrow$ Initial velocity

$a \rightarrow$ acceleration

$S \rightarrow$ displacement

$$(iv) \quad F = \frac{mv^2}{R}$$

(Physics formula)

PYQ (2023)

$$F \rightarrow \text{Force} = MLT^{-2}$$

$$m \rightarrow \text{mass} = M$$

$$v \rightarrow \text{velocity} = LT^{-1}$$

$$R \rightarrow \text{Radius} = L$$

So

$$MLT^{-2} = \frac{M(LT^{-1})^2}{L}$$

$$MLT^{-2} = \frac{ML^2T^{-2}}{L}$$

$$MLT^{-2} = MLT^{-2}$$

Q. Find out the Dimensions of a and b in Vander waal equation
(2024) JKBOSE

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

QUESTIONS

H/W

2022 (PYQ)



The velocity of a particle (v) at a instant (t) is given by $v = at + bt^2$ the dimension of b is:

- 1 L
- 2 LT^{-1}
- 3 LT^{-2}
- 4 LT^{-3}

SUMMARY



$$F = \frac{mv^2}{R}$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

C PO

Revision

DF

5
~
h
e

HOMework

H/w

Give the Dimensional formula of R (universal gas constt)

#

$$PV = nRT$$

$P \rightarrow$ Pressure

$V \rightarrow$ Volume

$n \rightarrow$ no. of mole

R

$T \rightarrow$ Temperature

Thank
You